

SUGIL S

AI / ML Engineer | Generative AI | Agentic AI | LLM Applications | RAG | Cloud AI & MLOps

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PROFESSIONAL SUMMARY

AI/ML Engineer with 2+ years of experience designing, building, and deploying production-grade AI systems across NLP, LLMs, and decision-support use cases. Skilled in end-to-end ML pipelines, retrieval-augmented generation (RAG), and real-time inference on distributed architectures. Experienced in shipping cloud-native AI services on AWS and Azure using Docker, Kubernetes, and modern MLOps practices, with a consistent record of cutting manual effort by 65-85% across enterprise workflows. Currently leading a small AI engineering team delivering banking and enterprise automation solutions.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

AI / Machine Learning: NLP, Large Language Models (LLMs), RAG, Prompt Engineering, Deep Learning, Neural Networks, Feature Engineering, Predictive Modeling, Anomaly Detection, Model Evaluation

AI Platforms & Models: AWS Bedrock, Azure OpenAI, OpenAI Whisper, Vision Models, Vector Databases

Cloud & MLOps: AWS (Bedrock, S3, Lambda), Azure, GCP, Docker, Kubernetes, Terraform, CI/CD, Azure DevOps, GitHub, GitLab, Bitbucket

Backend & Frontend: FastAPI, Flask, REST APIs, GraphQL, Microservices, React.js, Playwright

Data & Distributed Systems: Apache Kafka, Redis, Snowflake, PostgreSQL, Real-Time Streaming Pipelines

PROFESSIONAL EXPERIENCE

AI/ML Engineer — Team Lead — Xyant Services

Nov 2025 – Present

- Lead development of AI-driven decision-support and predictive systems for banking and enterprise automation, owning architecture, delivery, and technical mentoring for the AI team.
- Design end-to-end ML pipelines covering data ingestion, feature engineering, model training, evaluation, and production deployment.
- Build real-time AI inference services on Kafka-based distributed architectures supporting low-latency, high-throughput workloads.
- Deploy and scale containerized AI services using Docker and Kubernetes with automated CI/CD workflows.
- Drive architecture trade-off decisions across model accuracy, inference latency, and infrastructure cost.

Technical Analyst — VDart

Apr 2025 – Oct 2025

- Built a Call Analyzer AI platform with real-time audio processing, sentiment analysis, and filler-word detection, reducing manual call review effort by 80%.
- Developed a Mock Interview Simulation System with AI-driven feedback loops, improving training throughput and efficiency by 70%.
- Designed an LLM-powered candidate engagement platform that increased satisfaction scores by 65%.
- Created an AI-based email automation system that streamlined end-to-end recruitment communication workflows.

AI & ML Analyst — Avasoft

Dec 2023 – Mar 2025

- Developed enterprise AI chatbots on AWS Bedrock for natural-language querying and automated reporting across business systems.
- Built AI-driven analytics solutions for multi-cloud cost optimization across AWS and Azure environments.
- Engineered Kafka-based distributed pipelines for real-time data processing and Snowflake pipelines supporting large-scale AI workloads.

KEY PROJECTS

Real-Time Fraud Detection System (Decision AI)

AI/ML Engineer

Tech Stack: Python, Apache Kafka, Scikit-learn, Feature Engineering, Redis, Docker

- Built an anomaly detection model identifying fraudulent transaction patterns in high-volume streaming data, improving detection accuracy by 35% over rule-based baselines.

- Designed a Kafka-based real-time pipeline for continuous ingestion and online inference, sustaining sub-second decision latency at high throughput.
- Applied advanced feature engineering and predictive modelling to cut false-positive alerts by 40%, reducing analyst review workload.

AI-Powered Learning & Evaluation System

Full Stack & AI Developer

Tech Stack: Azure OpenAI, OpenAI Whisper, Python, Flask, React.js, PostgreSQL

- Developed an AI application that automates the trainee review process end-to-end using LLMs and speech recognition, reducing manual evaluation effort by 75%.
- Engineered a knowledge-base-driven question engine where the LLM generates syllabus-aligned questions and tracks intern activity throughout the review session.
- Integrated OpenAI Whisper for speech-to-text processing, achieving ~95% transcription accuracy across varied accents and audio quality.
- Implemented automated response validation with real-time feedback, cutting evaluation turnaround time from days to minutes (~85% faster).
- Delivered the full stack — Flask APIs, React.js interface, and PostgreSQL data layer — supporting concurrent multi-trainee review sessions.

AI-Powered Cloud Architecture Automation

Backend & AI Developer

Tech Stack: Python, React.js, AWS, Azure, GCP, Terraform, GitHub, GitLab, Azure DevOps, Bitbucket

- Automated cloud architecture design using AI driven by user inputs such as project name, cloud provider (AWS / Azure / GCP), and architecture pattern (DataOps, Serverless, and others), reducing design cycle time by 70%.
- Generated validated architecture diagrams with a human-in-the-loop regeneration flow, allowing architects to iterate in minutes instead of hours.
- Built AI-driven infrastructure-as-code generation with direct SCM integration across GitHub, GitLab, Azure DevOps, and Bitbucket, eliminating ~65% of manual Terraform authoring effort.
- Implemented per-service cost tracking dashboards that improved cloud spend visibility and enabled 20-25% reduction in avoidable infrastructure costs.

Report Generator using AWS Bedrock

Full Stack & AI Developer

Tech Stack: AWS Bedrock, Vision Models, Playwright, Python, Flask, Snowflake

- Developed an AI-powered report generator processing call transcripts, meeting documents, and cloud architecture artefacts, reducing report preparation effort by 80%.
- Implemented document chunking and context-management strategies enabling reliable summarisation of large files well beyond single-prompt context limits.
- Applied vision models to interpret business flow diagrams and auto-generate detailed narrative descriptions, removing a fully manual documentation step.
- Delivered AI-generated cloud architecture analysis covering strengths, risks, and recommended improvements for client review.
- Automated AWS billing extraction using Playwright, replacing ~90% of manual cost-tracking effort with a scheduled pipeline.

AI-Powered Insight Generation System

Full Stack & AI Developer

Tech Stack: AWS Bedrock, Snowflake, Python, Flask, React.js

- Built an automated insight engine that generates Snowflake queries via AWS Bedrock and converts live business data into narrative insights, reducing analyst reporting effort by 70%.
- Automated weekly, monthly, and quarterly insight cycles with scheduled email distribution to clients, removing recurring manual reporting work.
- Connected directly to Snowflake for real-time business data retrieval, keeping every insight current at generation time.
- Developed an AI chatbot enabling users to query client-specific data conversationally, cutting ad-hoc data request turnaround by 85%.

EDUCATION

Master of Computer Applications (MCA) — Bishop Heber College

2021 – 2023

Bachelor of Computer Applications (BCA) — Bishop Heber College

2018 – 2021